When AR Meets AI: A Duke Lab is Teaching Headsets to Build Worlds on Command

By Hadi Abdul

Imagine slipping on an AR headset, mumbling "uh, make me a campfire, something relaxing," and watching a flickering 3D fire materialize on the floor in front of you in under a minute. That's not science fiction. It's a Tuesday afternoon at the Duke I³T Lab, where one NCSSM senior is helping rewrite the rules of how virtual content gets made.

Joshua Chilukuri, a senior in the North Carolina School of Science and Mathematics Mentorship Program, spent the past school year embedded in Dr. Maria Gorlatova's Interactive, Intelligent, Internet of Things (I³T) Lab at Duke University, working alongside PhD candidate Yanming Xiu to evaluate how cutting-edge AI models can generate 3D objects in augmented reality on the fly, just from a person's voice. His work, conducted on a Meta Quest 3 headset connected to a beefy edge server packing three NVIDIA RTX 3090 GPUs, asks a deceptively simple question: when AI builds a 3D world for you, do you actually like what it makes?

"Our study demonstrates that Image-to-3D methods outperform text-to-3D for overall user satisfaction, despite higher latencies," Joshua explained. In plain English: people would rather wait a little longer for something that actually looks right.

The problem Joshua tackled is one that anyone who has ever tried to build an AR or VR experience knows intimately. Creating 3D models by hand takes hours, sometimes days. Searching online asset libraries can eat just as much time. It's the kind of bottleneck that keeps cool ideas trapped in someone's notebook. Joshua and his mentors set out to build a pipeline that could compress that process into a single sentence and a short wait.

The system they constructed is genuinely elegant. A user speaks into the AR headset, the audio gets streamed to an edge server, OpenAI's Whisper transcribes it, and a lightweight language model called Mistral-7B cleans up the "uhs" and "ums" into a tidy prompt. From there, the prompt either goes straight to a text-to-3D model, or takes a detour through Flux-Schnell to first generate a reference image, which then feeds into models like TripoSR or Microsoft's TRELLIS to produce the final 3D object. The whole thing pops up in the user's field of view, ready to be grabbed and moved around.

To figure out which of these four pipelines actually works best, Joshua helped run an IRB-approved user study with eleven participants, each asked to generate three objects: an apple, a wooden chair with a back, and a horse statue made of white stone. Participants rated the results on six different dimensions, from texture quality to whether the latency annoyed them. The winning pipeline, Flux paired with TRELLIS, scored 4.55 out of 5 for overall satisfaction on the horse statue task, even though it took the longest to generate at around 50 seconds. The fastest pipeline, Shap-E, cranked out objects in 20 seconds but scored a dismal 1.36 on the same task.

What makes this project stand out is not just the engineering. It's that, to the team's knowledge, this is the first work to bring Flux-Schnell, Microsoft TRELLIS, and TripoSR together inside an AR 3D content generation system. They also established a standardized set of metrics that future researchers can use to evaluate these systems in AR contexts.

Joshua's experience in the Mentorship Program offered him something most high school students never get: a seat at the table of active university research. He co-authored a paper, "Say It, See It," submitted to arXiv in August 2025, alongside Xiu, fellow NCSSM senior Shunav Sen, and Dr. Gorlatova. He's also a co-author on a second paper from the same lab, "A Neurosymbolic Framework for Interpretable Cognitive Attack Detection in Augmented Reality," which tackles a completely different but equally fascinating problem: detecting when AR overlays have been maliciously manipulated to deceive users.

The implications of his work stretch well beyond the lab. Teachers could generate 3D models of historical artifacts on demand for classroom demonstrations. Game designers and digital artists could prototype scenes in real time. Engineers could brainstorm CAD models out loud during meetings. Anywhere that 3D content creation is currently a bottleneck, this kind of pipeline could break it open.

There are still limits to push against. The team's user study was relatively small, the prompts were predetermined, and the system still depends on a beefy edge server rather than running fully on-device. But Joshua is clear-eyed about what the research established and what comes next. "Simpler object prompts that are well represented in text-to-3D datasets are preferred by users," he noted, a finding that points the way toward better training data and more efficient models down the line.

For now, the takeaway from Joshua's year at the I^{3}T Lab is both technical and human: AI can build worlds from words, but quality still beats speed in the eyes of the people who have to live in those worlds. As generative models keep evolving, and they are, fast, the question of what users actually want is one that researchers like Joshua are making sure stays at the center of the conversation.

"Image-to-3D methods outperform text-to-3D for overall user satisfaction," he said, summing up his work. The future of AR, it seems, is going to be spoken into existence, one well-formed prompt at a time.

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